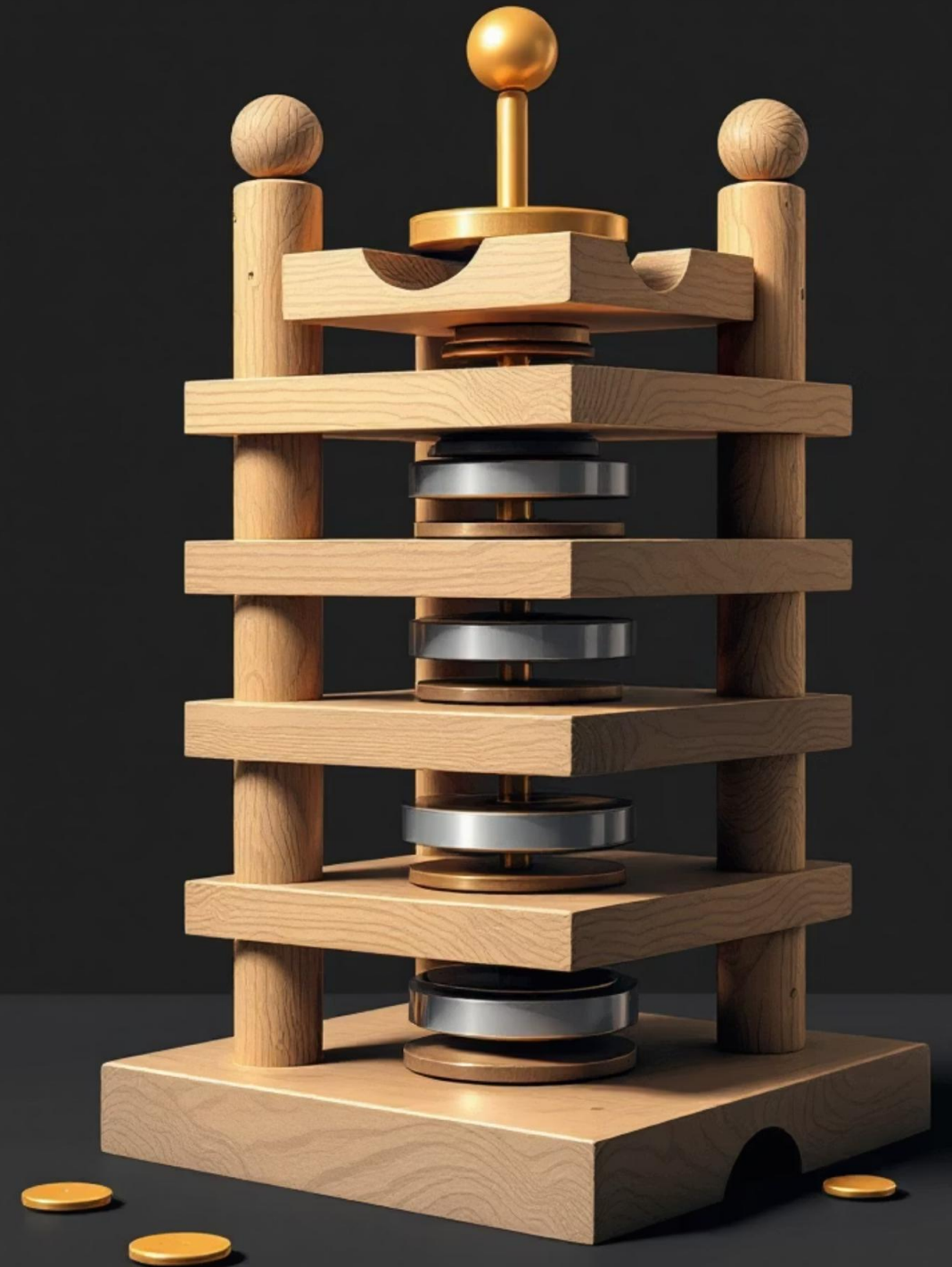


Tower of Hanoi

Algorithm & Recursion Analysis



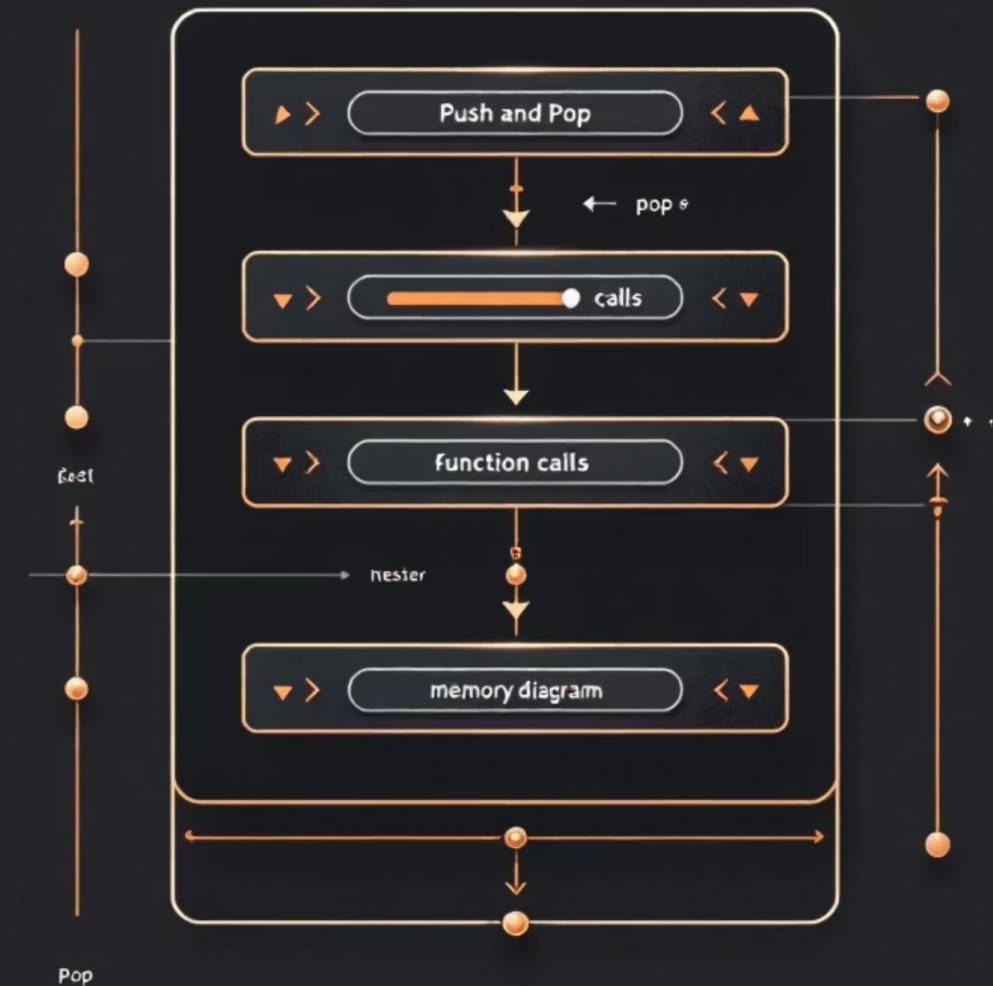
Pseudocode & Call Stack Trace

Recursive Function

```
hanoi(n, source, dest, aux):  
  1. if n == 1: move disk from source to dest else:  
  2. hanoi(n-1, source, aux, dest)  
  3. move disk from source to dest  
  4. hanoi(n-1, aux, dest, source)
```

Stack Trace (n=3)

Seven moves required.
Call stack grows to depth 3, tracking recursive calls and disk movements across pegs.





Complexity & Key Takeaways

Time Complexity

$$T(n) = 2T(n-1) + 1 \Rightarrow O(2^n)$$

Exponential growth: each disk doubles moves.

Space Complexity

$$O(n)$$

Call stack depth equals number of disks.

Learning Outcome

Recursion elegantly solves complex problems by breaking them into identical subproblems.